

Non-nutrient causes of low-grade, systemic inflammation: support for a 'canary in the mineshaft' view of obesity in chronic disease.

A form of low-grade, systemic inflammation ('metaflammation') is linked to many types of chronic disease. Initially, this was thought to be causally related to weight gain and obesity and a possible explanation of the link between obesity and disease. However, several lifestyle-related inducers of such inflammation, some of which are associated with obesity, but some of which are not, have now been identified. The most common of these have been nutritive related, suggesting that there could still be a relationship, either directly or indirectly, with obesity. Here we provide evidence for non-nutritive inflammatory inducers, providing further support for an earlier suggestion that while obesity, beyond a point, may have a direct link with disease, this may be neither necessary nor sufficient to explain the current epidemic of chronic disease. A more ubiquitous cause encompassing all inflammatory inducers is the modern, post-industrial environment and lifestyles emanating from this. Obesity may thus be more of 'a canary in the mineshaft', warning of bigger global problems, than just a single pathway to modern environmentally driven disease. © 2010 The Authors. obesity reviews © 2010 International Association for the Study of Obesity.